

9 September 2026

Editor-in-Chief

Computer Vision and Image Understanding

Dear Editor,

We submit the manuscript entitled “**A Paired Evaluation of Clean Accuracy and Corruption Sensitivity in Quantized Object Detection**” for consideration as a research article in *Computer Vision and Image Understanding*.

The paper addresses a measurement problem that arises when quantized object detectors are compared under degraded visual inputs. A raw corrupted-input accuracy gap combines any difference already present on clean images with the additional change associated with corruption. We therefore formulate an executable paired four-arm evaluation that reports clean treatment fidelity, remaining corrupted accuracy, and the change in the between-treatment gap under shared inputs and common image resampling.

The strongest experiment uses six selection-disjoint VOC/KITTI holdout blocks. FP8 exceeds INT8 by 1.60 AP points on original-clean images and by 1.05 AP points under corruption, while the clean-adjusted interaction is -0.55 AP with a paired 95% percentile interval of $[-0.78, -0.29]$. This result shows why absolute corrupted performance and corruption sensitivity can support different, non-contradictory interpretations. A broader 144-cell exploratory study across COCO, VOC, KITTI, and TT100K finds 56 conditions in which the raw corrupted gap and clean-adjusted interaction have opposite point signs. Controlled clean-input substitution, covariance audits, metric-scale sensitivity, seed/realization analyses, runtime measurements, and cross-family recipe-portability stress tests define the scope of the conclusion.

We believe the work fits CVIU because it concerns rigorous evaluation and understanding of deployed computer-vision systems under numerical and image-quality perturbations. The contribution is methodological and empirical: it provides an auditable comparison protocol for fixed quantized detector treatments and demonstrates the practical consequences of clean-baseline mixing, pairing, and absolute-accuracy reporting.

The manuscript is accompanied by Supplementary Information, reproducibility materials, highlights, and a graphical abstract. The authors retain full responsibility for the study and manuscript; use of generative-AI and AI-assisted tools is disclosed in the manuscript in accordance with Elsevier policy.

Thank you for considering our work.

Sincerely,

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